

Renewable Energy Development and Electricity System Performance in Africa

An Exploratory Analysis Using the CATF Electricity Sector Dataset

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Introduction

Overall Theme: Patterns and drivers of renewable energy development across African electricity systems

Research Goal: Investigate how renewable energy usage is different among African electricity systems and explore factors that may be associated with differences in renewable energy capacity between countries.

Target Population: The study focuses on African countries included in the CATF electricity sector dataset.

Audience: The primary audience is Clean Air Task Force (CATF) staff, who are interested in understanding patterns in electricity sector performance across African countries to help inform the development of the Electricity Sector Performance Scorecard (CATF-ESPS).

Data Summary

The dataset contains country-level indicators describing electricity system performance across African countries. These indicators capture areas such as electricity access, reliability, infrastructure development, financial performance, and renewable energy

Question #1: [Caellum Yip Hoi - Lee - Renewable Presence and Power Reliability]

Do countries with a higher share of renewable energy in the generation mix tend to experience more frequent power interruptions?

Q1: Variables & Approach

Variables used:

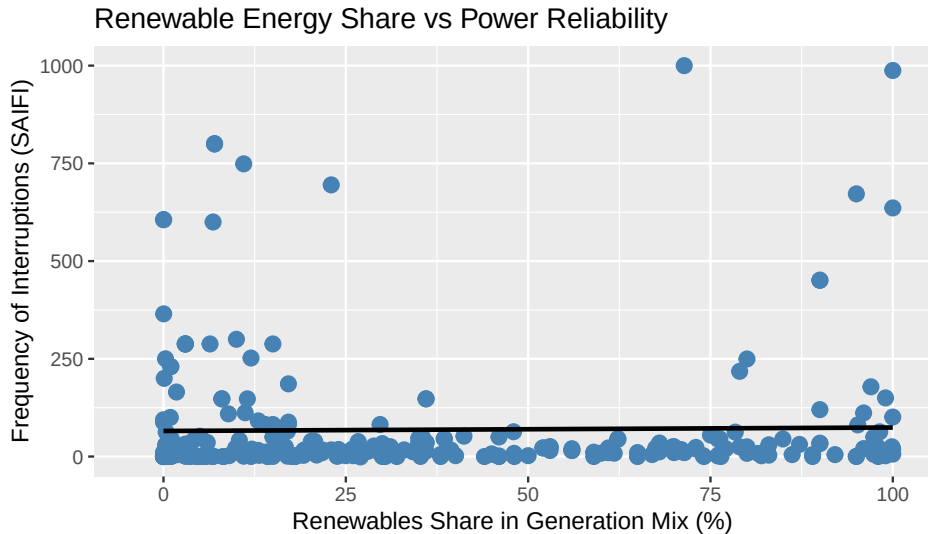
- **renewables_share_gen_mix_percent:** The percentage of electricity generation coming from renewable energy sources. This variable captures the amount of renewable energy being employed in the country's electricity system.
- **SAIFI:** System Average Interruption Frequency Index, representing Voltage Quality / Frequency It is the total number of reported interruptions divided by total customers. This variable serves as a primary measure of power reliability.

New variables/groups defined:

To intuitively compare reliability, I can group countries based on their renewable generation share:

- **renewable_group:** “Low” when `renewables_share_gen_mix_percent` is below 20%, “Medium” when it is between 20-50%, and “High” when above 50%

Q1: Visualization



Q1: Summary Table

Table 1: Average System Interruptions (SAIFI) by Renewable Energy Share

renewable_group	Countries	Avg Renewable Share (%)	Avg SAIFI (Interruptions)
Low (<20%)	219	6.2	82.9
Medium (20–50%)	110	32.7	31.0
High (>50%)	148	79.8	75.7

Q1: Insights & Takeaways

- **No clear linear relationship:** The scatter plot has a nearly flat trend line, suggesting that a higher share of renewables in the generation mix is not correlated with a higher frequency of power interruptions.
- **Reliability driven by other factors:** Countries with high interruption frequencies had widely varying levels of renewable energy in their generation mix. This suggests that issues with reliability and the subsequent SAIFI score likely has more to do with other factors, i.e infrastructure.
- **Renewables not sacrificing stability:** I see this evidence as positive, since it suggests that African electricity systems can have higher percentages of renewable energy without directly inducing a lower electric grid reliability.

Question #2: [Paanini Kartik - Development and Renewable Energy Usage]

Do more developed countries tend to generate a higher or lower proportion of renewable energy?

Q2: Variables & Approach

Variables from data:

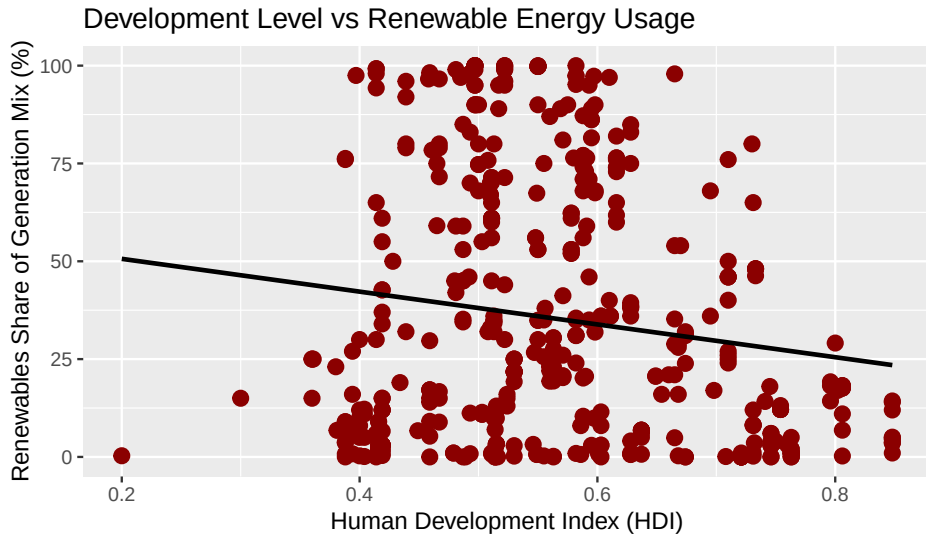
- **hdi** The Human Development Index of a given country. This will be used as a metric for how developed the country is.
- **renewables_share_gen_mix_percent** What % of the country's generation comes from renewables. We use this instead of `total_installed_renewables_capacity` as that may trivially become higher with more development.

Created variables:

In order to group countries based on their HDI to facilitate better comparison, we will use the HDI standard, as detailed by the UNDP.

- **hdi_grouping** “Low” when the HDI is less than 0.5. “Medium” when it is ≥ 0.5 and < 0.7 . “High” when it is ≥ 0.7 and < 0.8 . “Very high” when it is ≥ 0.8 .

Q2: Visualization



Q2: Summary Table

Table 2: Renewable Energy Share by HDI Development Group

hdi_grouping	Countries	Avg HDI	Avg Renewable Share (%)
Low (<0.5)	144	0.43	35.3
Medium (0.5–0.7)	240	0.57	43.1
High (0.7–0.8)	68	0.74	15.1
Very High (≥ 0.8)	20	0.82	11.7

Q2: Insights & Takeaways

- Visually speaking, there appears to be a slight negative correlation between the two variables, given the line of best fit having a slightly negative slope.
- However, there appears to be a large spread in the data, with data points not clustering around said line. Moreover, there are less countries with higher HDI values than lower ones, as apparent by the summary table.
- As a result, further hypothesis testing soon to be done will be able to give a stronger claim for whether or not a negative correlation truly exists.

Question #3: [Hyman - Private Sector Participation and Renewable Energy Adoption]

Is private sector participation in power generation associated with renewable energy adoption in African countries?

Q3: Variables & Approach

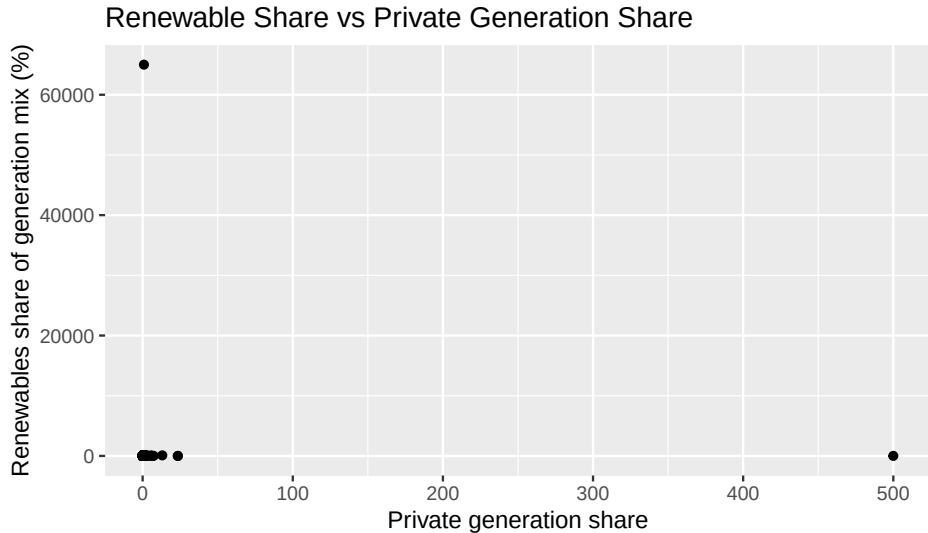
Variables used:

- `private_sector_power_gen_commercial`
- `renewables_share_gen_mix_percent`
- `installed_gen_capacity`

New variables/groups defined:

- `private_generation_share <- private_sector_power_gen_commercial / installed_gen_capacity`

Q3: Visualization



Q3: Summary Table

```
# A tibble: 2 x 4
  private_group      n mean_renewable median_renewable
  <chr>          <int>         <dbl>         <dbl>
1 Higher           72      930.           20.6
2 Lower          224      35.6           20.6
```


- There is no clear and strong relationship between private sector participation in power generation and renewable usage.
- Countries with lower private participation have a variety of renewable outcomes, whilst countries with higher private participation did not have higher renewable outcomes.
- The median across both higher and lower groups was the same, further suggesting that renewable usage may not be explained solely by the private sectors role in power generation.

Question #4: [Sahand Samadirad - Revenue Collection and Renewable Capacity]

Do countries with higher electricity revenue collection rates tend to have more installed renewable energy capacity?

Q4: Variables & Approach

Variables used:

- **collection_rate_percent**

Percentage of billed electricity revenue that utilities successfully collect.

This variable reflects the financial efficiency and operational performance of electricity utilities.

- **total_installed_variable_renewables_capacity**

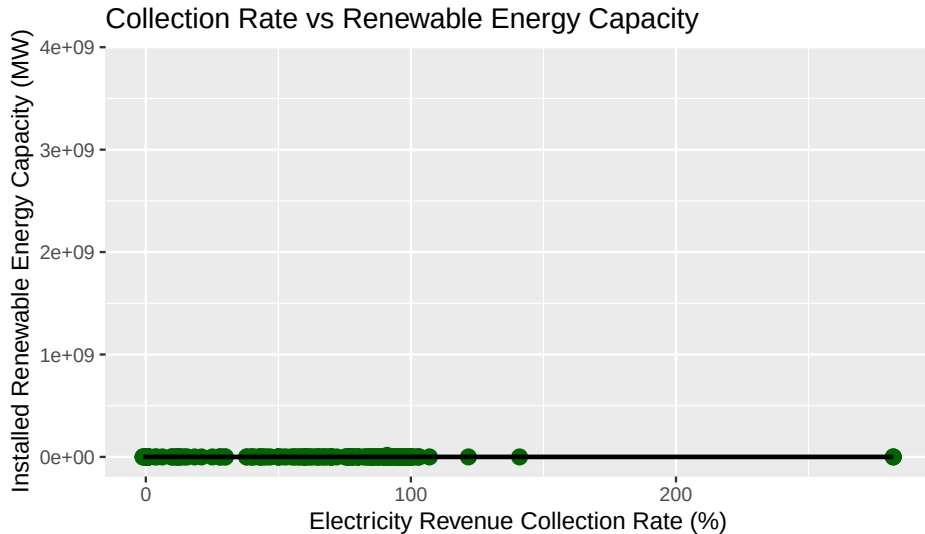
Total installed capacity of variable renewable energy sources such as wind and solar (MW).

This variable represents the level of renewable energy development within a country's electricity system.

New variables/groups defined:

To make comparisons clearer, countries are grouped based on electricity revenue collection performance:

Q4: Visualization



Q4: Summary Table

Table 3: Average Renewable Capacity by Collection Rate Category

collection_group	Countries	Avg Collection Rate (%)	Avg Renewable Capacity (MW)
High (>90%)	403	100.9	13081007.2
Low (<70%)	119	28.6	21282.8
Medium (70–90%)	63	80.2	432.4

Overall Limitations

Data Limitations:

- **Data Collection:** The CATF data was collected by student internet searches. This introduces lots of room for inaccuracies, since data points may come from different years or use different definitions.
- **Inaccuracies in data:** Outliers such as the 65 000% renewables_share_gen_mix indicate that the data set contains some inaccurate data that will impact the results of the research.
- **Missing Data:** Many African countries may lack the necessary infrastructure to report all the variables in the dataset. This may skew the results of research; it may cause the less wealthy countries to be less represented in the dataset.

Future Directions

- **Standardized Data Collection:** To reduce inaccuracies introduced by varying methods of data collection, the CATF dataset could be collected only from standardized global databases, i.e World Bank. This would minimize inconsistent value but may lead to more values being dropped.
- **Data Cleansing:** To improve the removal of extreme outliers (like the 65,000% renewable share), strict checks could be implemented during data entry to automatically remove physically impossible values before analysis. For example, a check could be implemented to ensure that the values inputted in the `renewables_share_gen_mix` are never greater than 100.
- **Inclusive Data Collection Methods:** Instead of collecting data directly from the government records, globally available proxy data could be used to ensure that every country is included. While it is unfeasible to manually collect the data for each country, the CATF could use stand-in values as estimates. For example, the amount of imported solar panels could be used as a rough proxy for solar capacity.